Ayush Agarwal

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Summary

AI & ML Specialist with expertise in architecting Generative AI applications, RAG-based systems, and end-to-end predictive pipelines. Proven track record of deploying LLM-powered tools (Gemini, LangChain) and high-accuracy machine learning models to automate complex decision-making. Specializes in bridging rigorous data science methodologies with production-ready AI solutions to accelerate operational efficiency.

Skills

Languages & Databases: Python, SQL, Snowflake
AI & Big Data: LangChain, scikit-learn, XGBoost, PySpark, Prophet, pandas, numpy
Tools & Cloud: Databricks, AWS, Docker, Streamlit Cloud, Git/GitHub
Core Competencies: Generative AI, RAG Architectures, Predictive Modeling, Model Deployment

Certifications & Training

- **Snowflake Data Cloud:** Completed foundational technical pathways and earned official data warehousing learner badges.
- **Artificial Intelligence using TensorFlow:** SmartInternz & Google.

Experience

Labcorp Bengaluru, India
Data Analyst Associate Intern Feb 2026 – Aug 2026

- Engineered automated data extraction workflows on large-scale datasets using Python and SQL to identify critical anomalies and complex patterns.
- Developed technical data visualizations to validate data integrity, reducing manual pre-processing time for downstream analytical modeling.
- **Tools Used:** Python, SQL (Snowflake), Pandas, NumPy, Scikit-learn, PowerBI

Research & Publications

Predictive Modelling of Physicochemical Properties (QSPR & QSAR) *Letters in Applied NanoBioScience*

- Authored multiple research manuscripts focused on predicting chemical bioactivity using Topological Indices, deploying ensemble regression models (Random Forest, AdaBoost) to optimize computational accuracy.
- **Published QSPR Paper:** [Link to Paper / DOI: 10.33263/LIANBS143.192]

Urban Plant Suitability Index (UPSI) Framework *Environmental and Sustainability Indicators*

- Designed and computed a novel multi-criteria decision-making framework to evaluate tropical urban ecosystems by synthesizing diverse environmental variables into a single statistical scoring metric.
- **Published Paper:** [Link to Paper]

Core Projects

QuickBite Express – High-Performance Optimization Engine *Python, Streamlit, Scikit-learn*

- Engineered an intelligent operational optimization engine utilizing machine learning to streamline delivery dispatching and routing decisions.
- Implemented predictive models to accurately estimate demand surges and dynamic constraints, improving overall processing efficiency.

InsightIQ – Agentic Text-to-SQL LLM Engine *LangChain, Gemini API, Vector DB, Python*

- Architected a RAG-based Generative AI agent that translates natural language questions into complex, executable SQL queries with self-correction loops.
- Integrated vector databases to store metadata schemas, enabling autonomous data retrieval and reducing insight extraction latency for non-technical users.

SCM-Optima – AI Supply Chain Control Tower *Prophet, Random Forest, Python, Streamlit*

- Developed a predictive supply chain analytics platform using Random Forest and Prophet for 30-day demand forecasting and shipment delay risks.
- Built an interactive Streamlit application displaying real-time risk scores, enabling preemptive logistical adjustments with an **83%** model accuracy.

Education

Your University Name Bengaluru, India
Bachelor of Technology (B.Tech.) in Your Engineering Branch Aug 2022 – June 2026
Cumulative GPA: **8.69/10.0**